

2. Comparison with Specialized Models											
		MAE/MSE ↓					F1-SEDI 99.5/99 ↑				
	task	Temperature	Dewpoint	Wind Direc. ($\times 10^3$)	Wind Rate	Sea-Level Pressure	Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure
iTransformer	forecast 48h→72h	3.29/21.98	3.54/25.91	77.83/9.84	1.70/6.17	5.18/58.49	0.15/0.52	0.1/0.48	↓	0.01/0.10	0.13/0.37
Conformer		3/17.72	3.16/30.88	76.56/9.74	1.59/4.44	4.30/40.91	0.17/0.56	0.28/0.51	↓	0.04/0.12	0.15/0.40
Ours		1.62/6.94	1.62/7.43	66.8/8.86	1.35/4.36	1.17/4.06	0.50/0.79	0.44/0.75	↓	0.24/0.43	0.76/0.86
SSSD	imputation	5.49/35.21	4.51/28.33	116.82/16.46	2.52/8.85	5.67/47.54	0.22/0.51	0.26/0.59	↓	0.17/0.27	0.31/0.57
Ours		0.91/1.89	0.99/2.12	49.0/6.01	1.09/2.09	0.99/9.61	0.74/0.89	0.69/0.86	↓	0.26/0.53	0.99/0.99
Diffusion-TS	generation	4.23/33.06	3.66/30.44	88.58/13.59	1.94/7.89	4.22/38.68	0.13/0.42	0.19/0.54	↓	0.15/0.25	0.21/0.49
Ours		1.79/7.27	1.82/7.93	71.9/9.50	1.43/4.47	1.27/4.57	0.45/0.76	0.36/0.71	↓	0.14/0.33	0.75/0.85

3. Parameter Sensitivity Analysis																							
UniGLO Version			MAE/MSE ↓					F1-SEDI 99.5/99 ↑					FD ↓					STE ↓					Cross ↓
	param	scope	Temperature	Dewpoint	Wind Direc. ($\times 10^3$)	Wind Rate	Sea-Level Pressure	Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure	Temperature	Dewpoint	Wind Direc.	Wind Rate	Sea-Level Pressure	Temperature	Dewpoint	Wind Direc. ($\times 10^3$)	Wind Rate	Sea-Level Pressure	
8 expert	3.73 MB	136.18 GFLOPs	1.58/5.82	1.58/6.33	66.98/8.75	1.33/3.82	1.12/3.53	0.54/0.80	0.46/0.75	↓	0.17/0.39	0.79/0.88	0.24	0.25	0.57	0.39	0.17	2.58	1.51	2.53	1.04	0.85	1.49
16 expert	4.01 MB	136.38 GFLOPs	1.59/5.86	1.58/6.30	69.32/8.73	1.33/3.85	1.10/3.49	0.52/0.79	0.45/0.74	↓	0.16/0.39	0.78/0.88	0.24	0.25	0.56	0.38	0.17	2.57	1.5	2.51	1.03	0.84	1.49
32 expert	4.57 MB	137.53 GFLOPs	1.53/5.61	1.56/6.25	67.72/8.69	1.33/3.78	1.04/3.24	0.55/0.80	0.46/0.75	↓	0.19/0.39	0.79/0.88	0.23	0.24	0.55	0.37	0.15	2.55	1.49	2.47	1	0.82	1.47
1M	0.99 MB	39.19 GFLOPs	1.65/5.79	1.61/6.36	68.97/8.59	1.34/3.80	1.18/3.61	0.52/0.79	0.45/0.74	↓	0.17/0.38	0.75/0.87	0.25	0.27	0.58	0.44	0.19	2.7	1.54	2.61	1.09	0.9	1.5
15M	14.63 MB	501.90 GFLOPs	1.50/5.39	1.54/6.09	66.31/8.65	1.31/3.74	1.01/3.02	0.55/0.80	0.47/0.75	↓	0.21/0.42	0.80/0.89	0.22	0.24	0.53	0.33	0.14	2.5	1.45	2.36	0.97	0.76	1.44